

## Multiple Choice (10 marks)

1. A water pipe in which water pressure is  $80 \text{ lb/in.}^2$  springs a leak, the size of which is  $.01 \text{ sq in.}$  in area. How much force is required of a patch to "stop" the leak?
  - (a) 8 oz
  - (b) 0.008 lb
  - (c) 8 kg
  - (d) 800 lb
  - (e) .8 lb
2. A nucleus of mass  $M$  emits a  $\gamma$  ray of energy  $E_{\gamma \text{ref}}$ . The nucleus was initially at rest. What is
  - (a)  $E_{\gamma}^2 / 4Mc^2$
  - (b)  $2E_{\gamma}^2 / Mc^2$
  - (c)  $E_{\gamma} / Mc^2$
  - (d)  $3E_{\gamma}^2 / 2Mc^2$
  - (e)  $E_{\gamma} / 4Mc^2$
3. When 100 J are put into a device that puts out 40 J of useful work, the efficiency of the device is
  - (a) 25%
  - (b) 100%
  - (c) 60%
  - (d) 50%
  - (e) 75%
4. Find the minimum energy of a simple harmonic oscillator using the uncertainty principle.
  - (a)  $\frac{h\omega}{4\pi}$
  - (b)  $m\omega^2/2$
  - (c)  $\frac{h}{m\omega^2}$
  - (d)  $\frac{h^2}{2\pi m\omega}$
  - (e)  $h^2/m\omega$
5. What is the total charge in coulombs of  $75.0 \text{ kg}$  of electrons?
  - (a)  $-2.05 \times 10^{13} \text{ C}$

- (b)  $-1.32 \times 10^{13} \text{C}$
- (c)  $-1.98 \times 10^{13} \text{C}$
- (d)  $-0.98 \times 10^{13} \text{C}$
- (e)  $-1.44 \times 10^{13} \text{C}$

## True / False (10 marks)

*Answer True or False*

6. True or False: An electron can reach a speed of  $1.8 \times 10^7 \text{ m/s}$  by being accelerated across a potential difference of 100 V.
7. True or False: A uniformly charged conducting sphere with a diameter of 1.2 m and surface charge density of  $2 \times 10^{-6} \text{ C/m}^2$  has a total charge of  $2.5 \times 10^{-5} \text{ C}$ .
8. True or False: An atom with filled  $n = 1$  and  $n = 2$  energy levels contains a total of 12 electrons.
9. True or False: The focal length of the lens is -20 inches, indicating that it is a diverging lens.
10. True or False: The specific heat capacity of sand is always equal to that of vegetation, leading to the same temperature change for the same amount of heat added.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Analyze the formation of the image by a bi-convex lens with a radius of curvature of 20 cm when an object of height 2 cm is placed 30 cm away. Discuss the characteristics of the image produced.
12. An observer moves past a meter stick at a velocity of half the speed of light. Discuss how relativistic effects influence the measured length of the meter stick and calculate the observed length.

*End of Paper*